Chapter 5 Root Locus Method

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§ 1 Introduction

1.1 Direct Analysis Method

—Time Domain Solution

- **Best for judging performance**
- **Too complicated**
- **Difficult to predict the performance as parameters change**
- **Time consuming**

1.2 Requirements of Analysis Methods

- **Simple to implement**
- **Easy to predict the system performance**
- **Possible to indicate better parameters**

1.3 Major Indirect Methods

(1) Algebraic Stability Criterion

- **Routh Stability Criterion**
  - ✶ **Absolute stability**
  - ✶ **No proper measure of relative stability**
  - ✶ **Difficult to judge that if the CL poles reside in a given region**
  - ✶ **No clue to parameter changes**

(2) Frequency Response Method

- **Nyquist plot, Bode diagrams, Nichols charts are available**
- **Easy to predict the CL performances**
- **Easy to improve the CL performance by modifying OL frequency response**
- **Not able to know the relation between the CL poles and the OL parameters**

(3) Root Locus Method

- **What is a root locus?**

  A plot of the Roots of

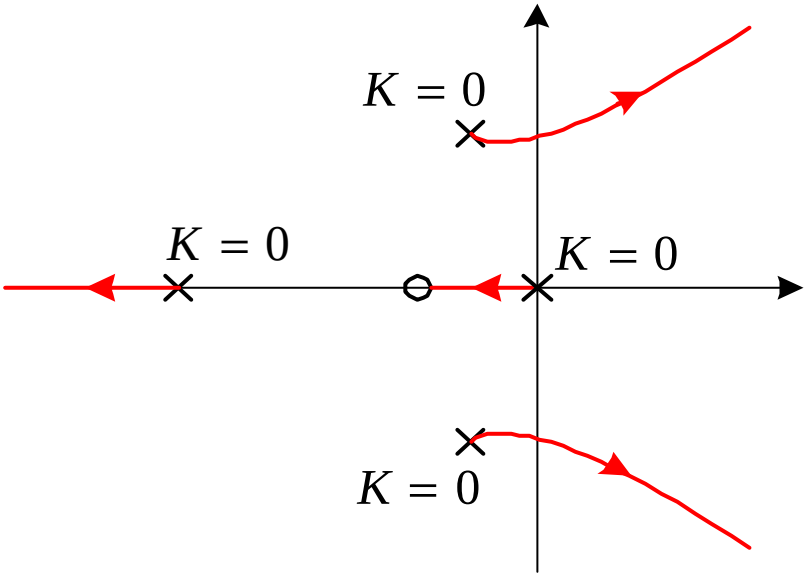
  of the characteristic equation

  of the Closed Loop

  as the function of Gain

- **Why to study the root locus?**

  Easy to predict the CL performances



- **Possible to study the root locus?**

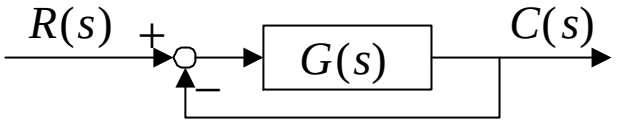
  YES. W R Evans, 1948, 1950

§ 2 Root Locus Plots

2.1 An Example

Example 2.1 Draw the CL root locus for the given OL transfer function

$$G(s)=\frac{K}{s(s+1)}$$



Solution: CL TF is

$$G_{\mathrm{CL}}(s)=\frac{C(s)}{R(s)}=\frac{G(s)}{1+G(s)}=\frac{K}{s^2+s+K}$$

K **varies → CL poles change.**

CL characteristic equation:

$$s^2+s+K=0$$

CL poles are:

$$s_{1,2}=-\frac{1}{2}\pm\frac{1}{2}\sqrt{1-4K}$$

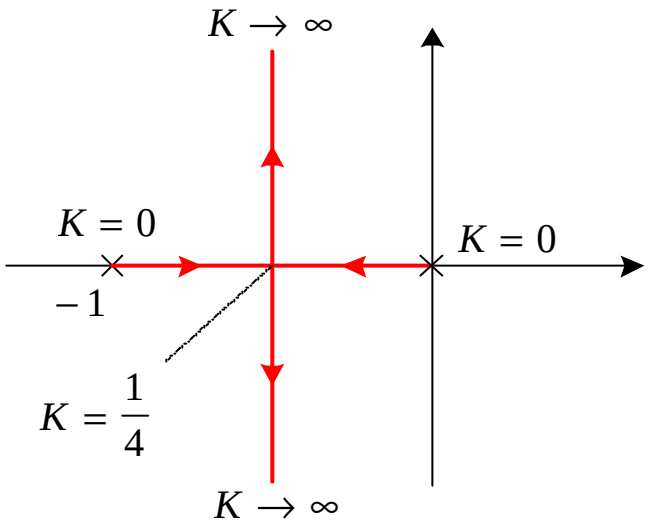
(i) $K\geq 0$

$K=0$**,** $s_{1,2}=-\frac{1}{2}\pm\frac{1}{2}=0,\ -1$**,** **OL poles**

$$K < \frac{1}{4},\ s_{1,2} = -\frac{1}{2} \pm \frac{\sqrt{1-4K}}{2}$$

$$K = \frac{1}{4},\ s_{1,2} = -\frac{1}{2} \pm 0 = -\frac{1}{2},\ -\frac{1}{2}$$

$$K > \frac{1}{4},\ s_{1,2} = -\frac{1}{2} \pm \mathrm{j}\frac{\sqrt{4K-1}}{2}$$



CL system damping

K=0 0.25 ¥

Overdamped Underdamped

(ii) $K \le 0$

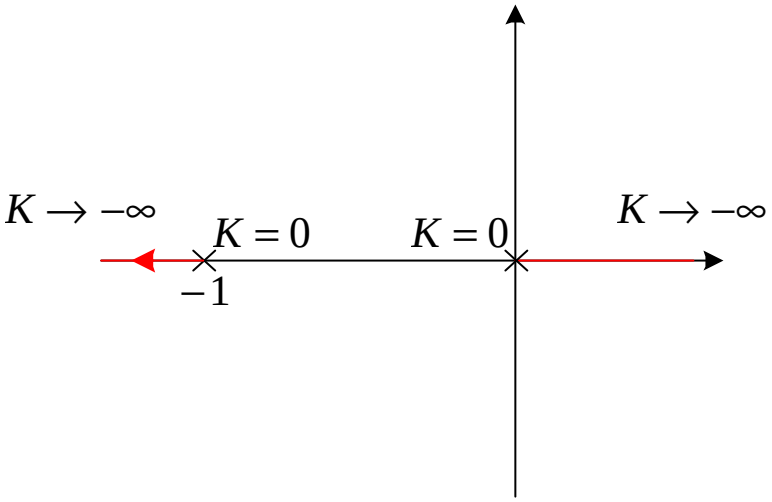
$K = 0,\ s_{1,2} = 0,\ -1,$ **OL poles**

$K < 0$

$$s_{1,2} = -\frac{1}{2} \pm \frac{\sqrt{1+4|K|}}{2}$$

$s_1 \in (-\infty, -1]$

$s_2 \in [0, +\infty)$



□

2.2 Angle and Magnitude Conditions

- **Assumption:** $K \geq 0$
- **Root locus conditions**

All CL poles satisfy

$$G(s)F(s) = -1$$

i.e.

Magnitude condition: $|G(s)F(s)| = 1$

Angle condition:

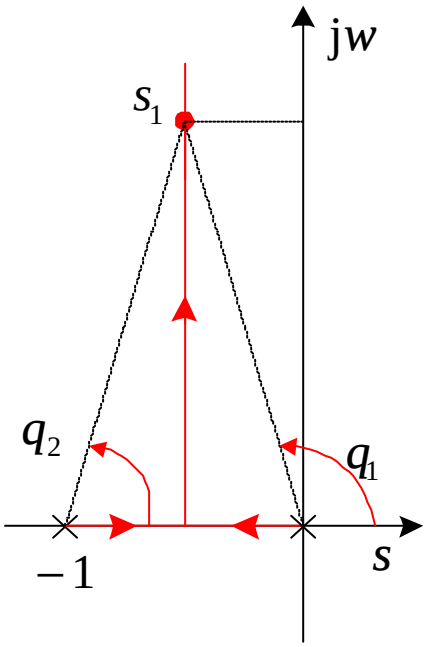
$$\arg[G(s)F(s)] = \pm 180^\circ(2k+1)$$

$$k = 0,1,2,\cdots$$

Example 2.2 Condition check for Example 2.1

Solution: Take arbitrarily $s_1 = -\frac{1}{2} + \mathrm{j}2$

(i) Angle condition



$$\arg G(s_1) = \arg\frac{1}{s_1(s_1+1)}$$

$$= -\arg(s_1) - \arg(s_1+1)$$

$$= -q_1 - q_2$$

$$= -\arctan\frac{2}{-0.5} - \arctan\frac{2}{0.5}$$

$$= -104.04^\circ - 75.96^\circ = -180^\circ$$

(2) Magnitude condition

$$|G(-0.5 \pm \mathrm{j}2)| = \frac{K}{|-0.5+\mathrm{j}2|\ |-0.5+\mathrm{j}2+1|}$$

$$= \frac{4K}{17} = 1$$

i.e. $K = \frac{17}{4}$ Þ **CL poles** $s_1 = -\frac{1}{2} \pm j2$ □

- **Angle condition ALONE gives root locus**

  Ý

  All CL poles satisfy angle condition without knowing K
- **Magnitude condition gives** K **for a given CL pole**

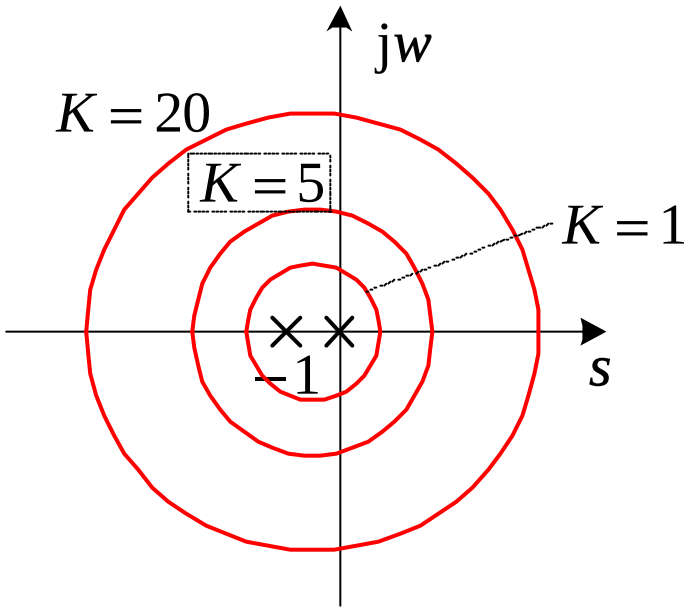
2.3 Constant Gain Locus

- **Locus of complex** s **satisfying** $|G(s)F(s)| = 1$ **with constant** K
- **Magnitude condition ALONE gives constant gain locus**

Example 2.3 Draw the constant gain locus for

$$G(s)F(s) = \frac{K}{s(s+1)}$$

Solution:



□

- **Intersecting points of root locus and constant gain locus are the CL poles**

- **Root locus and constant gain locus are always orthogonal**

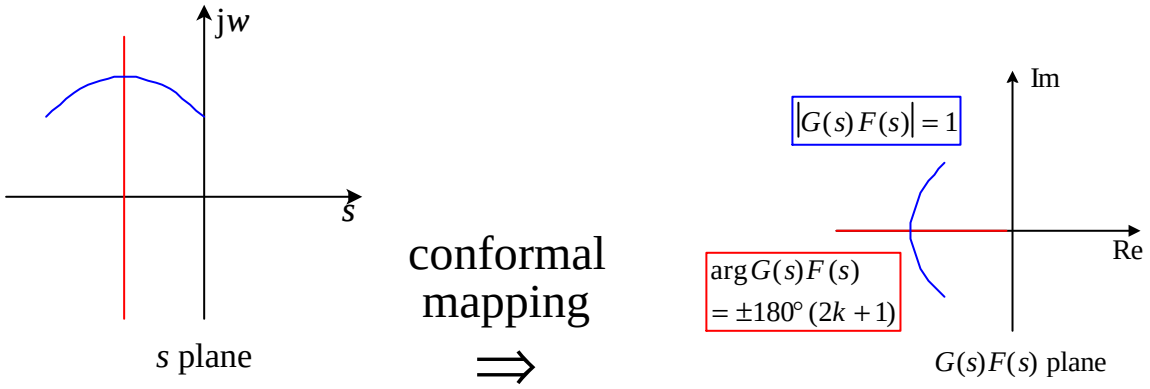
In $G(s)F(s)$ plane

★ $\arg[G(s)F(s)] = \pm 180°(2k+1)$ **and** $|G(s)F(s)| = 1$ **are conformal mappings of root locus and constant gain locus**

★ $\arg[G(s)F(s)] = \pm 180°(2k+1)$ **and** $|G(s)F(s)| = 1$ **are orthogonal**

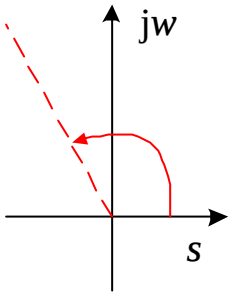
So in s plane

Root locus and constant gain locus are orthogonal



§ 3 General rules for Constructing Root Locus

- **Assumptions:**
  - ✶ **Argument:**



Positive real axis—0°

counterclockwise

— +ve direction

  - ✶ **Negative F/B—** $K \geq 0$
  - ✶ **Only CL root locus on the upper** s **plane are calculated**

3.1 Write Char. Eqn. in the Form

$$1+G(s)F(s)=1+KW(s)=1+K\frac{B(s)}{A(s)}$$

$$=1+\frac{K(s-z_1)\cdots(s-z_m)}{(s-p_1)(s-p_2)\cdots(s-p_n)}=0$$

And LOCATE the OL poles and zeros in the s **plane.**

3.2 Find Starting/Terminating Points and Number of Branches

- **Starting points:** $(K = 0)$

$$|KW(s)| = 1 \quad \text{Þ} \quad |W(s)| = \frac{1}{K}$$

$$\lim_{K \to 0} |W(s)| = \lim_{K \to 0} \frac{1}{K} = \infty$$

It is equivalent to

$$(s - p_1)(s - p_2) \cdots (s - p_n) = 0$$

✶ **All OL poles are starting points**

- **Terminating points:** $(K = \infty)$

$$\lim_{K \to \infty} |W(s)| = \lim_{K \to \infty} \frac{1}{K} = 0$$

It is equivalent to

$$\begin{cases} (s - z_1)(s - z_2) \cdots (s - z_m) = 0 \\ s \to \infty \qquad \text{if} \quad n > m \end{cases}$$

✶ **All OL zeros are (finite) terminating points**

✶ **If** $n > m$**, there are** $n - m$ **infinite terminating points**

- **Number of Branches**

n **branches** ¬ n **starting points**

where m **branches to** m **finite zeros**

$n - m$ **branches to infinity**

***NB*: If** $n < m$**,**

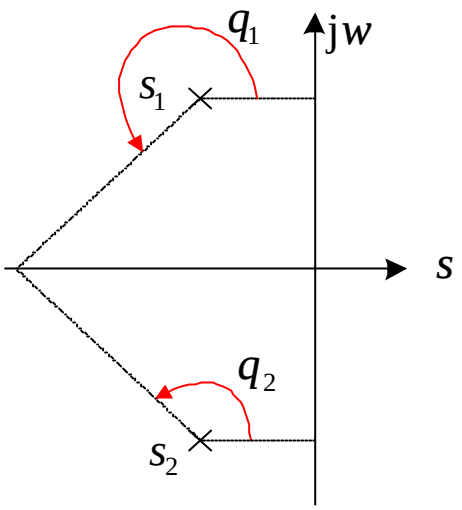
n **branches from finite OL poles**

$m - n$ **branches from infinity**

All branches terminate at OL zeros

3.3 Loci on the Real Axis

- **Depends only on the real poles & zeros**
- **Conjugate poles make no contributions to the angle condition**



$$q_1 + q_2 = 360°$$

- **Odd number of real poles and zeros to the right of loci**
  - ✶ **Real axis point to the right of poles/zeros**





Phase angle ° 0°

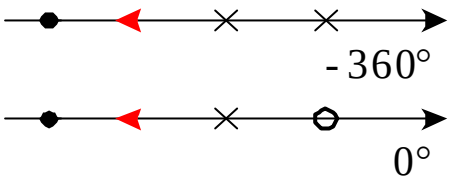
  - ✶ **Real axis point to the left of poles/zeros**





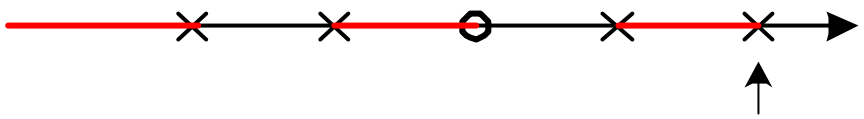
Phase angle ° 180°

✶**If** $2k$ **poles/zeros on the right to a real axis point**



Phase angle$= \pm 180° \times 2k$ **which does not satisfy Angle Condition**

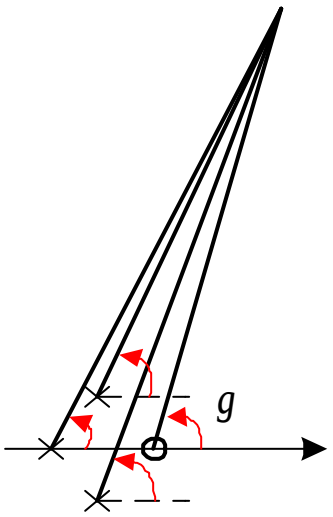
✶**Draw real axis root loci**



Rightest pole / zero

3.4 Asymptotes of loci as $s \to \infty$

●**Angle of asymptotes**



As $s \to \infty$

All phase angle are the same—g

$$\arg[G(s)F(s)]\big|_{s\to\infty} = (m-n)g$$

From $\arg[G(s)F(s)] = \pm 180°(2k+1)$

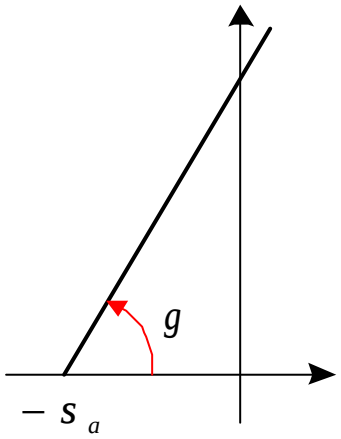
we have $g = \dfrac{\mp 180°(2k+1)}{n-m}, \quad k = 0,1,2,\cdots$

- $n-m$ **distinct asymptotes**

Angle of one asymptote with the real axis

$$g = \frac{-180°(2k+1)}{n-m}, \quad k = 0,1,\cdots,n-m-1$$

- **Real axis intersect of asymptotes**

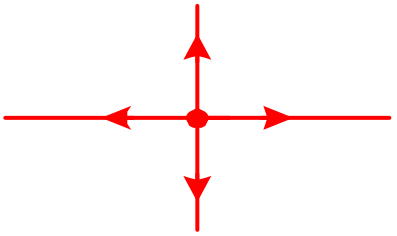


$$-s_a = \frac{\sum_{j=1}^{n} p_j - \sum_{i=1}^{n} z_i}{n-m}$$

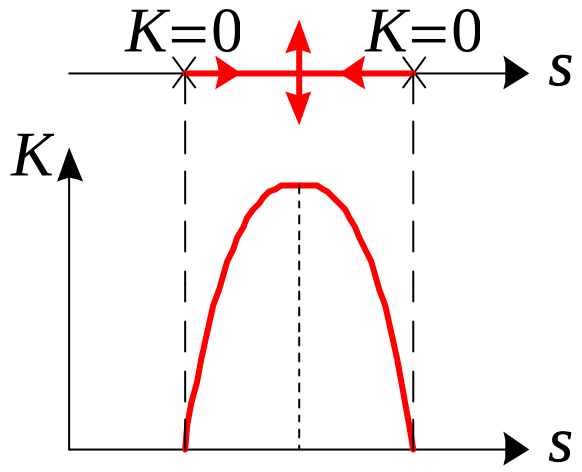
(See Appendix I for the proof)

3.5 Breakaway and break-in points

- **Multiple CL poles case**

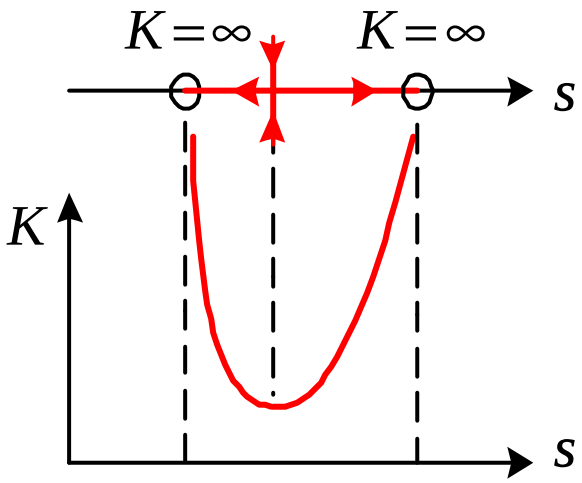


- **Real axis breakaway/break-in points**



Break-in point

K **takes a maximum**



Breakaway point

K **takes a minimum**

***NB*: (i) There may be more than ONE breakaway/break-in points**

(ii) Breakaway/break-in points are not necessarily on the real axis

- **Calculation**

CL poles satisfy

$$f(s) = A(s) + KB(s) = 0 \qquad \textbf{(1)}$$

Multiple CL poles satisfy (at least)

$$\frac{\mathrm{d}f(s)}{\mathrm{d}s} = A'(s) + KB'(s) = 0$$

$$K = -\frac{A'(s)}{B'(s)} \qquad \textbf{(2)}$$

Substituting Eqn (2) into Eqn (1) gives

$$A(s) - \frac{A'(s)}{B'(s)}B(s) = 0$$

i.e $A(s)B'(s) - A'(s)B(s) = 0$ **(3)**

Since $K = -\dfrac{A(s)}{B(s)}$**,**

so $\dfrac{\mathrm{d}K}{\mathrm{d}s} = \dfrac{A(s)B'(s) - A'(s)B(s)}{B^2(s)}$ **(4)**

Comparing Eqns (3) and (4) gives the necessary condition:

$$\frac{dK}{ds} = 0 \qquad \textbf{(5)}$$

***NB*: Breakaway/break-in points are only those points satisfying** $K > 0$

Example 3.1 Given

$$G(s)F(s) = \frac{K}{s(s+1)(s+2)}$$

To find the breakaway/break-in points.

Solution: The characteristic equation is

$$K = -s(s+1)(s+2) = -s^3 - 3s^2 - 2s$$

$$\frac{dK}{ds} = 0 \;\; \textcircled{R} \;\; -3s^2 - 6s - 2 = 0$$

$$s = -0.423,\ -1.577$$

For $s = -0.423$

$$K = -s(s+1)(s+2)\Big|_{s=-0.423} = 0.385 > 0$$

For $s = -1.577$

$$K = -s(s+1)(s+2)\Big|_{s=-1.577} = -0.385 < 0$$

So -0.423 is the breakaway point.

***NB*:**

(i) -1.557 is not on the root locus,

Þ it is not a breakaway/

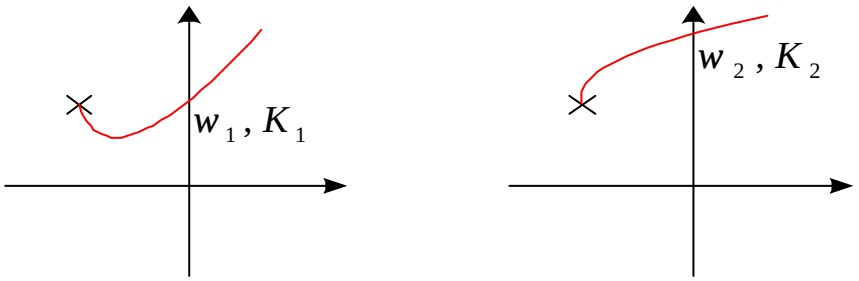
break-in point.

(ii) When $B(s) = 1$**, the necessary condition is** $A'(s) = 0$**.** □

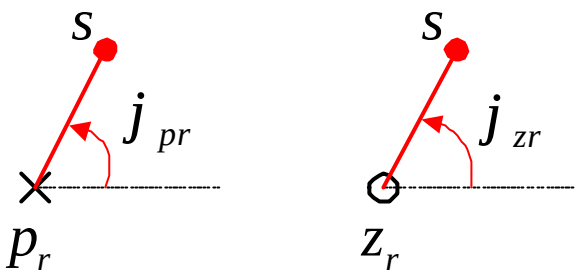
(See Appendix II for further discussions on multiple breakaway/break-in points)

3.6 Angle of Departure from Complex poles and Angle of Arrival at Complex Zeros

- **These angles are important for drawing**



- **A test point near the complex pole/zero is used**



- **Angle of departure from a complex pole p_r**

A test point s should satisfy

$$\arg G(s)F(s) = \sum_{i=1}^{m} \arg(p_r - z_i) - j_{pr} - \sum_{\substack{j=1 \\ j \neq r}}^{n} \arg(p_r - p_j)$$

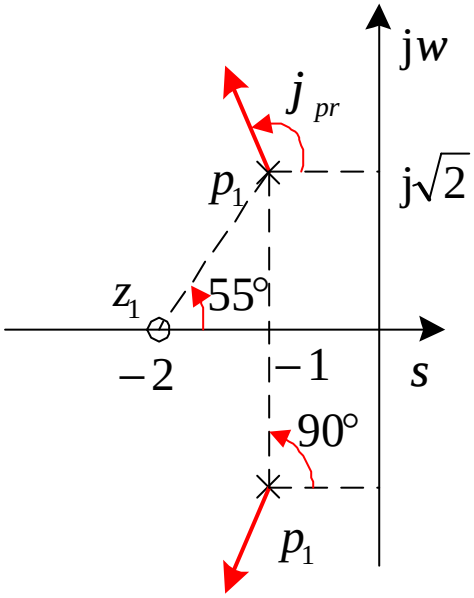
$$= \pm 180^\circ (2k+1)$$

$$j_{pr} = \pm 180^\circ (2k+1) - \sum_{\substack{j=1 \\ j \neq r}}^{n} \arg(p_r - p_j) + \sum_{i=1}^{m} \arg(p_r - z_i)$$

Example 3.2 Find the angle of departure for

$$G(s)F(s)=\frac{K(s+2)}{s^2+2s+3}$$

Solution: OL poles are $p_{1,2}=-1\pm \mathrm{j}\sqrt{2}$



$$j_{p1}=\pm 180°(2k+1)$$
$$-\arg(p_1-p_2)$$
$$+\arg(p_1-z_1)$$
$$=\pm 180°(2k+1)$$
$$-90°+55°$$
$$=145°$$

$j_{p2}=-145°$ **(symmetry)** □

- **Angle of arrival at a complex zero** z_r

A test point s **should satisfy**

$$\arg G(s)F(s) = \sum_{\substack{i=1 \\ i\neq r}}^{m} \arg(z_r - z_i) + j_{zr}$$

$$-\sum_{j=1}^{n} \arg(z_r - p_j) = \pm 180^\circ (2k+1)$$

$$j_{zr} = \pm 180^\circ (2k+1) + \sum_{j=1}^{n} \arg(z_r - p_j) - \sum_{\substack{i=1 \\ i\neq r}}^{m} \arg(z_r - z_i)$$

(See Appendix III for the angle of departure from multiple OL poles)

3.7 Imaginary Axis Crossing Points

- **Crossing of imag. axis by root locus**
  - Þ **change of stability status**
- **The crossing points indicate critical K and oscillation frequency**

- **Methods**

(i) Routh stability criterion

(ii) Let $s = \mathrm{j}w$**. Solve**

$$\begin{cases} \mathrm{Re}[1 + G(\mathrm{j}w)F(\mathrm{j}w)] = 0 \\ \mathrm{Im}[1 + G(\mathrm{j}w)F(\mathrm{j}w)] = 0 \end{cases}$$

for w and K**.**

(iii) Trial and error approach

Example 3.3 Find the crossing points for

$$G(s)F(s) = \frac{K}{s(s+1)(s+2)}$$

Solution:

Char eqn: $f(s) = s^3 + 3s^2 + 2s + K$

(i) Routh criterion

Routhian array

s^3	**1**	**2**	$6-K=0$
s^2	**3**	K	$K=6$
s^1	$(6-K)/3$	**Aux. Eqn from line** s^2	
s^0	K	$3s^2+K=0$	
		$s=\pm j\sqrt{2}$	

(ii) $f(jw)=(jw)^3+3(jw)^2+2(jw)+K=0$

$$\begin{cases} K-3w^2=0 \\ 2w-w^3=0 \end{cases} \qquad \begin{cases} w=\pm\sqrt{2} \\ K=3w^2=6 \end{cases}$$ □

3.8 Conservation of the Sum of System Poles

- **Sum of CL poles = Sum of OL poles if**

$n-m\geq 2$

$$f(s)=A(s)+KB(s)$$

$$=s^n-\sum_{i=1}^{n}p_i s^{n-1}+\cdots$$

$$= K\left[s^m - \sum_{j=1}^{n} z_j s^{m-1} + \cdots \right]$$

When $n - m \geq 2$, i.e. $m \leq n-2$.

$$f(s) = s^n - \sum_{i=1}^{n} p_i s^{n-1}$$

$$+ \sum \text{terms of } (s^{n-2}, s^{n-3}, \cdots, s^m, s^{m-1}, \cdots)$$

Suppose the CL poles are $l_1, l_2, \cdots, l_n$,

then $f(s) = \prod_{i=1}^{n} (s - l_i)$

$$= s^n - \sum_{i=1}^{n} l_i \, s^{n-1} + \cdots$$

So $\sum_{i=1}^{n} l_i = \sum_{i=1}^{n} p_i$

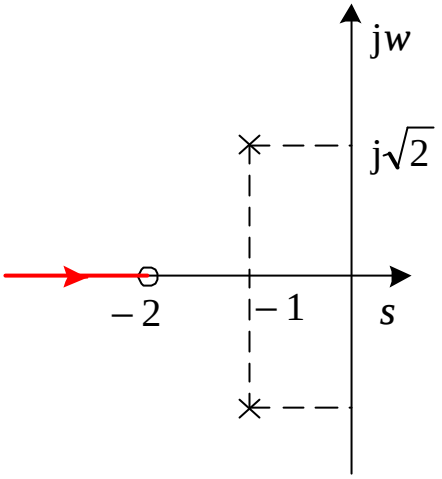
- **Part of CL poles may be found according to the rule**

3.9 Illustrative Examples

Example 3.4 Sketch the root locus for

$$G(s)=\frac{K(s+2)}{s^2+2s+3},\quad F(s)=1$$

Solution:



(i) $G(s)F(s)=\dfrac{K(s+2)}{s^2+2s+3}$

OL poles: $p_{1,2}=-1\pm \mathrm{j}\sqrt{2}$

OL zeros: $z_1=-2$

Char eqn: $f(s)=s^2+2s+3+K(s+2)$

$$=s^2+(2+K)s+(3+2K)=0$$

$$K=-\frac{A(s)}{B(s)}=-\frac{s^2+2s+3}{s+2}$$

(ii) Starting points: $-1\pm \mathrm{j}\sqrt{2}$

Terminating points: $-2,\ \infty$

Number of branches: 2

(iii) Real axis locus: $(-\infty,-2)$

(iv) Asymptotes

$$g = \frac{\pm 180°(2k+1)}{n-m} = \frac{\pm 180°(2k+1)}{1}$$

$$= -180° \qquad (k = 0)$$

No need to calculate $-s_a$

(v) Breakaway points

$$\frac{dK}{ds} = 0 \qquad \frac{A'(s)}{B'(s)} = \frac{A(s)}{B(s)}$$

$$\frac{2s+2}{1} = \frac{s^2+2s+3}{s+2}$$

$$s^2 + 4s + 1 = 0$$

$$s_{1,2} = -2 \pm \sqrt{3} = -3.732,\ -0.268$$

-3.732 **is on the root locus and is the breakaway point**

NB: $K = -\left.\frac{s^2+2s+3}{s+2}\right|_{s=-3.732} = 5.4641 > 0$

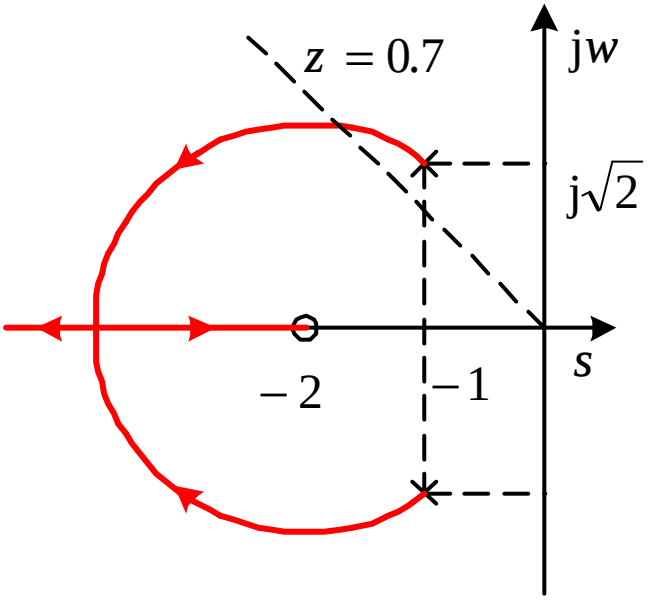
(vi) Angle of departure

$$j_{p1} = \pm 180° - \arg(p_1 - p_2) + \arg(p_1 - z_1)$$

$$= \pm 180° - 90° + 45° = 145° \quad \textbf{or} \quad -215°$$

$$j_{p2} = -145°$$

(vii) Imaginary axis crossing points



No crossing point

(viii) A simple application: To find the CL poles and corresponding gain for $z = 0.7$.

Let $s = -zw_n \pm \mathrm{j}w_n\sqrt{1-z^2}$

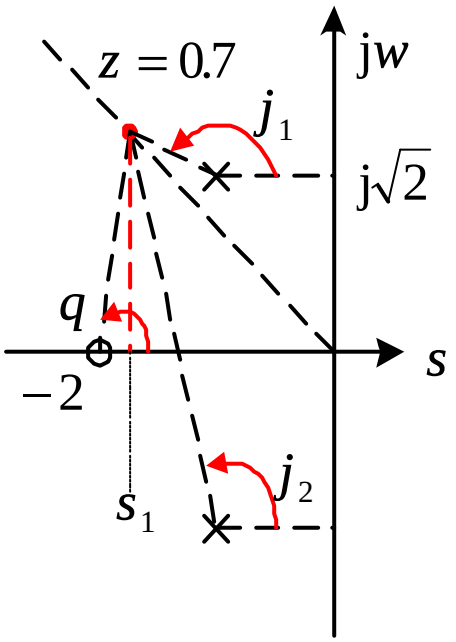
$$= -s \pm \mathrm{j}s\frac{\sqrt{1-z^2}}{z}$$

$$= -s \pm \mathrm{j}1.0202s$$

then $G(s) = \dfrac{K(-s + \mathrm{j}1.0202s + 2)}{(-s + \mathrm{j}1.0202s + 1 + \mathrm{j}\sqrt{2})(-s + \mathrm{j}1.0202s + 1 - \mathrm{j}\sqrt{2})}$

$$\arg G(s)F(s) = \arctan\frac{1.0202s}{2-s} - \arctan\frac{1.0202s + \sqrt{2}}{1-s}$$

$$- \arctan\frac{1.0202s - \sqrt{2}}{1-s}$$



$$= q - j_1 - j_2 = -180°$$

$$q + 180° = j_1 + j_2$$

$$\tan q = \frac{\tan j_1 + \tan j_2}{1 - \tan j_1 \tan j_2}$$

$$2.0408s^2 - 4s + 1 = 0$$

$$s_1 = 1.666, \quad s_2 = 0.294$$

Take $s_1 = 1.666$ **(See root locus for the reason)**

then $s = -1.67 \pm \mathrm{j}1.70$

According to magnitude condition

$$K = \left| \frac{(s+1)^2 + 2}{s+2} \right|_{s=-1.67 \pm j1.70} = 1.34$$

$$K = 1.34 \quad s_d = -1.67 \pm j1.70 \quad z = 0.7$$

(See Appendix IV for more discussions)

□

Example 3.5 Sketch the root locus for

$$G(s)F(s) = \frac{K}{s(s+2.73)(s^2+2s+2)}$$

Solution:

(i) OL poles: $0,\ -2.73,\ -1 \pm j$

$$K = -s(s+2.73)(s^2+2s+2)$$

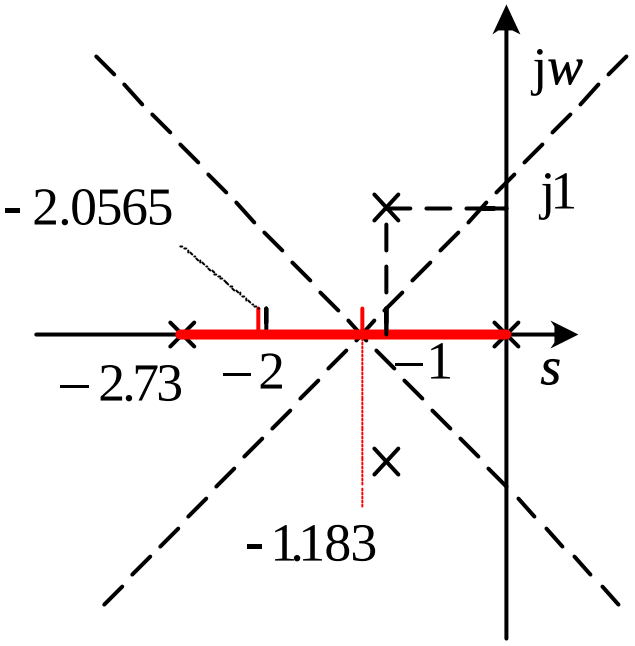
$$= -(s^4 + 4.73s^3 + 7.46s^2 + 5.46s)$$

(ii) Starting points: $0,\ -2.73,\ -1 \pm j$

Terminating points: μ

Number of branches: 4

(iii) Real axis locus: (-2.73, 0)



(iv) Asymptotes

$$g = \frac{\pm 180^\circ (2k+1)}{n-m} = \frac{\pm 180^\circ (2k+1)}{4}$$

$$= \pm 45^\circ,\ \pm 135^\circ$$

$$-s_a = \frac{\sum_{j=1}^{n} p_j - \sum_{i=1}^{m} z_i}{n-m}$$

$$= \frac{0 - 2.73 - 1 + \mathbf{j1} - 1 - \mathbf{j1}}{4}$$

$$= -1.183$$

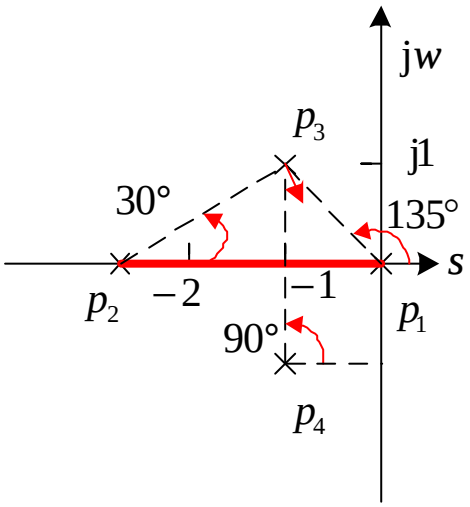
(v) Break-in points

$$\frac{dK}{ds} = 0, \quad 4s^3 + 14.19s^2 + 14.92s + 5.46 = 0$$

One real root $s = -2.0565$ **(on the root locus)**

$$K = -A(s)\big|_{s=-2.0565} = 2.931 > 0$$

(vi) Angle of departure



$$\begin{aligned} j_{p3} &= 180° - \arg(p_3 - p_1) \\ &\quad - \arg(p_3 - p_2) \\ &\quad - \arg(p_3 - p_4) \\ &= 180° - 135° - 30° - 90° \\ &= -75° \end{aligned}$$

$$j_{p4} = 75°$$

(vii) Imaginary crossing points

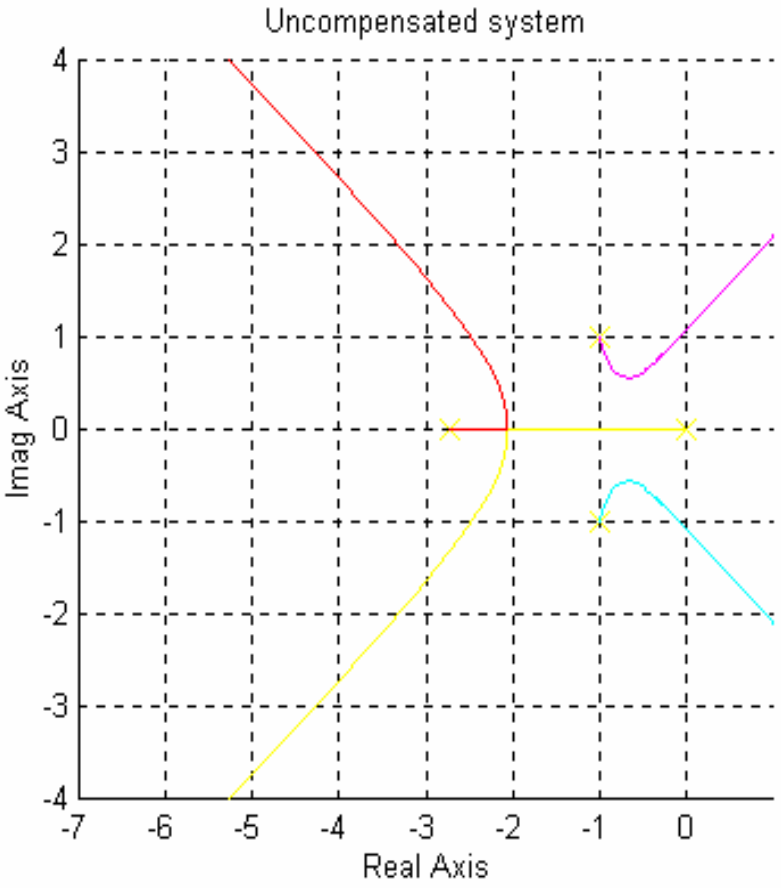
$$f(s) = s^4 + 4.73s^3 + 7.46s^2 + 5.46s + K = 0$$

Let $s = jw$

$$w^4 - j4.73w^3 - 7.46w^2 + j5.46w + K = 0$$

$$\begin{cases} K + w^4 - 7.46w^2 = 0 \\ 5.46w - 4.73w^3 = 0 \end{cases}$$

$$w = \pm 1.0744, \quad K = 7.28$$



□

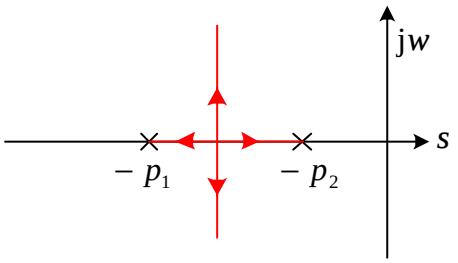
§ 4 Root Locus Analysis of Control Systems

4.1 Effect of Zeros and Poles

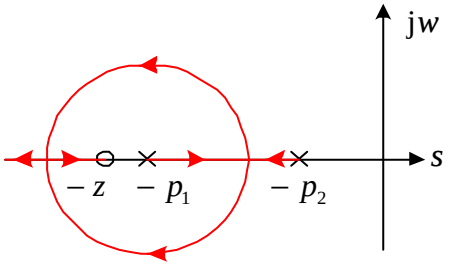
(1) Addition of Zeros and Poles

Example 4.1 Addition of a zero

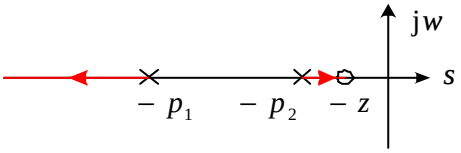
$$G(s)F(s) = \frac{K}{(s+p_1)(s+p_2)}$$



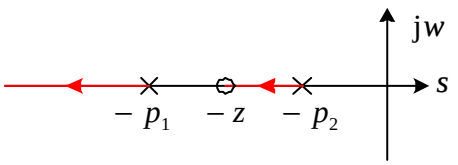
$p_1 > p_2 > 0$



$z > p_1 > p_2$



$p_1 > p_2 > z$

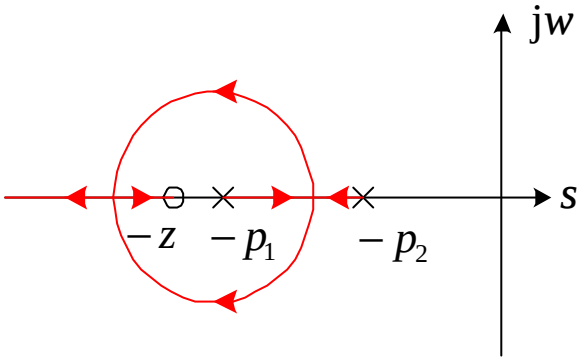


$p_1 > z > p_2$

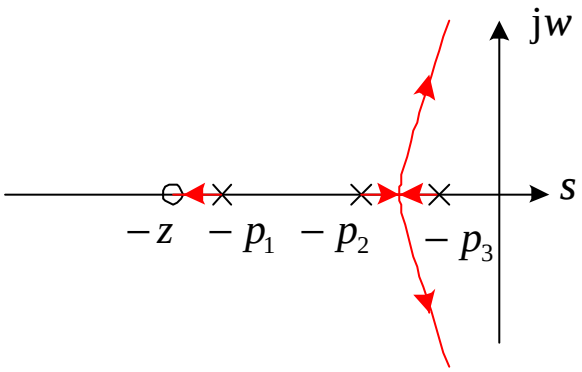
★ **A zero moves the root locus to the left**

Example 4.2 Addition of a pole

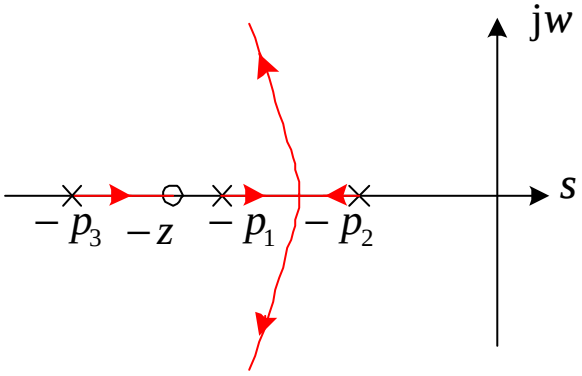
$$G(s)F(s) = \frac{K(s+z)}{(s+p_1)(s+p_2)}$$



$z > p_1 > p_2$



$p_3 < p_2$



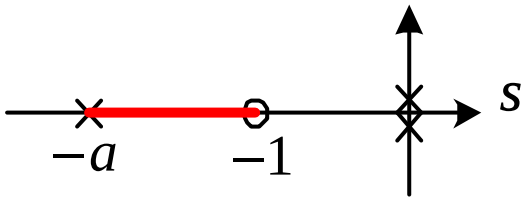
$p_3 > z$

★**A pole moves the root locus to the right**

(2) Movements of Poles and Zeros with a Parameter

Example 4.3 Given OL transfer function

$$G(s)F(s) = \frac{K(s+1)}{s^2(s+\alpha)}$$



It is important to check if there are break-in/breakaway points in $(-\alpha,-1)$

$$K = -\frac{A(s)}{B(s)} = -\frac{s^2(s+\alpha)}{s+1}$$

$$\frac{dK}{ds} = 0 \quad \text{Þ} \quad A(s)B'(s) = A'(s)B(s)$$

$$s^2(s+\alpha) = (3s^2 + 2\alpha s)(s+1)$$

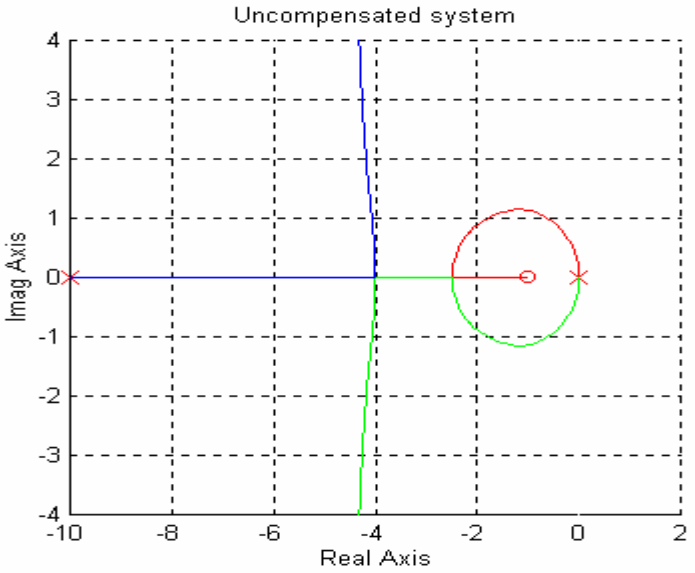
$$2s^2 + (\alpha+3)s + 2\alpha = 0$$

$$s = \frac{-(\alpha+3) \pm \sqrt{\alpha^2 - 10\alpha + 9}}{4}$$

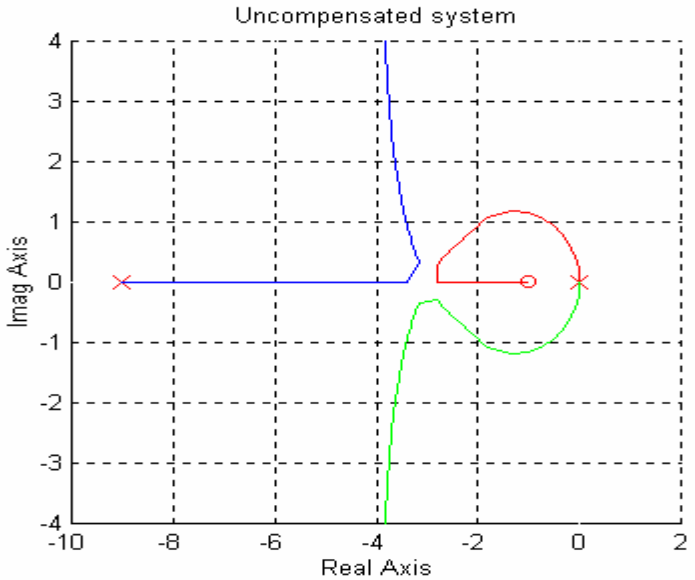
Break-in/breakaway points exist if

$$a^2 - 10a + 9 \geq 0, \quad \textbf{i.e.} \quad a \geq 9, \quad a \leq 1$$

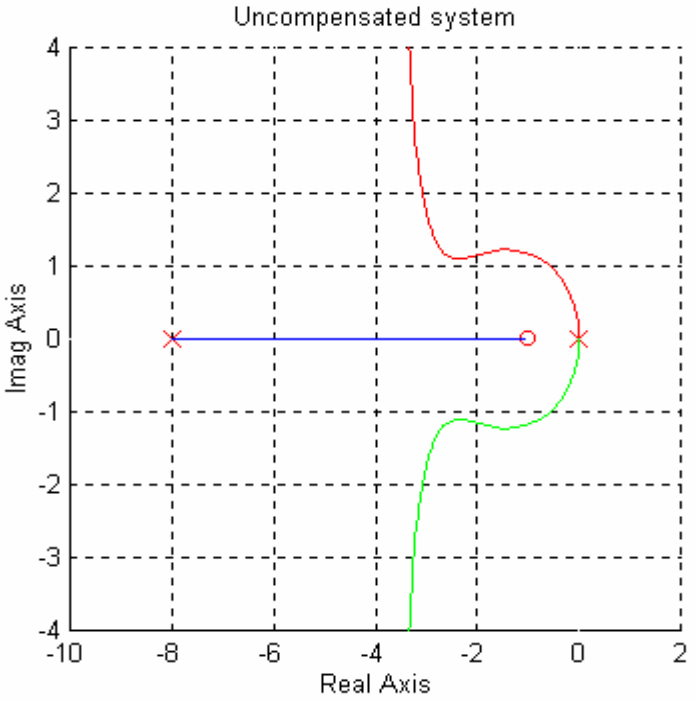
- $a = 10 \geq 9 \quad s_{1,2} = -4, \ -2.5$



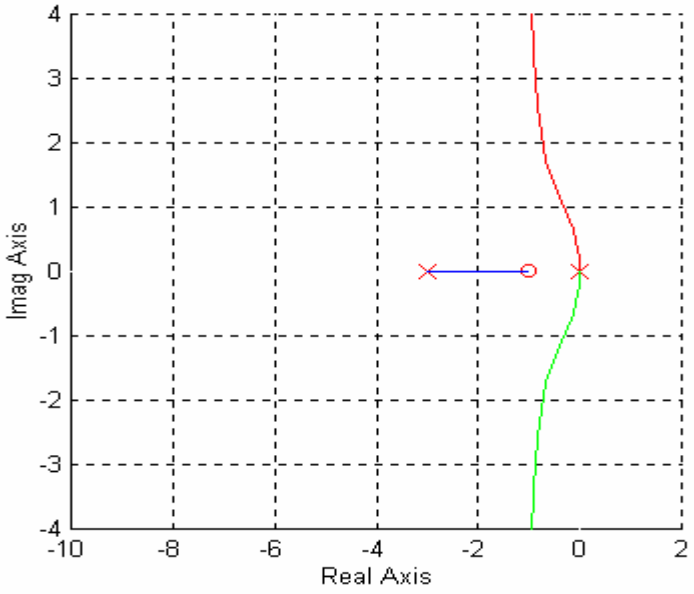
- $a = 9 \quad s_{1,2} = -3, \ -3$



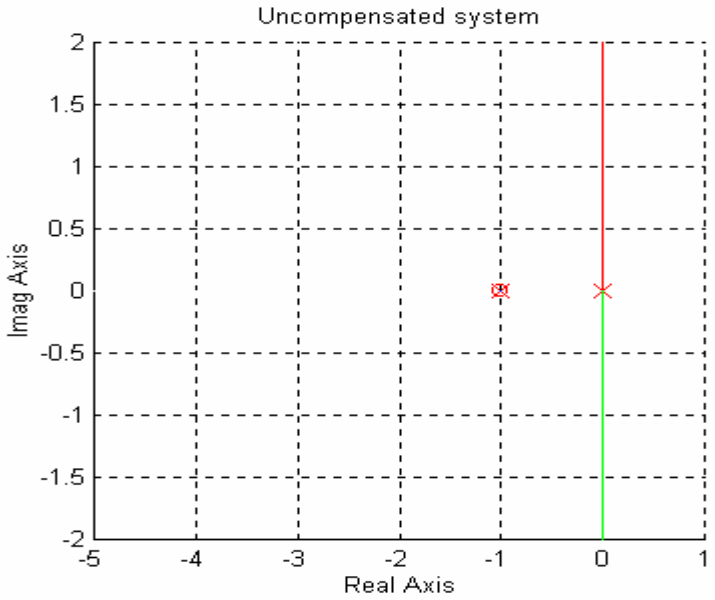
- $a = 8$ **No break-in & breakaway points**



- $a = 3$

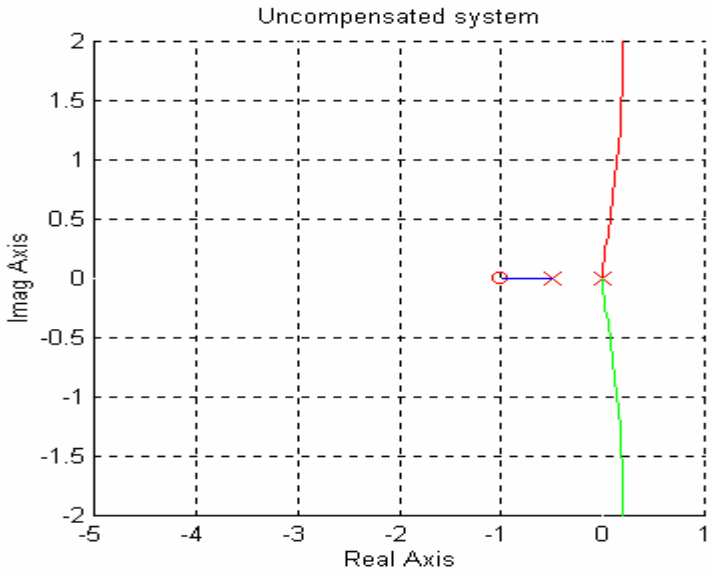


- $a = 1$ $\qquad G(s)F(s) = \dfrac{K}{s^2}$



***NB*: -1 must be a closed-loop pole.**

- $a = 0.5$



★Minor change in the OL poles may cause dramatic change in the CL root loci

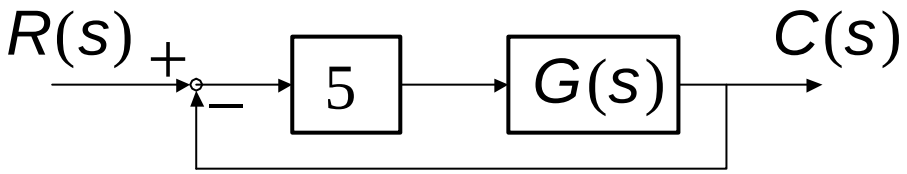
4.2 Comparison Study:

Example 4.4 Derivative control and velocity feedback in a positional servemechanism

$$G(s)=\frac{1}{s(5s+1)}$$

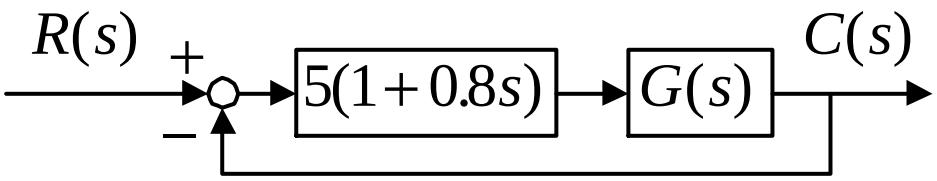
(i) Three Loop Configurations

- **P-control**



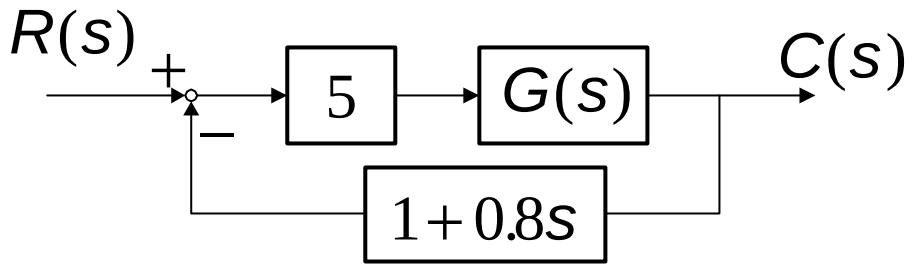
$$G_{\mathrm{I}}(s)F_{\mathrm{I}}(s)=\frac{5}{s(5s+1)}=\frac{1}{s(s+0.2)}$$

- **PD-control**



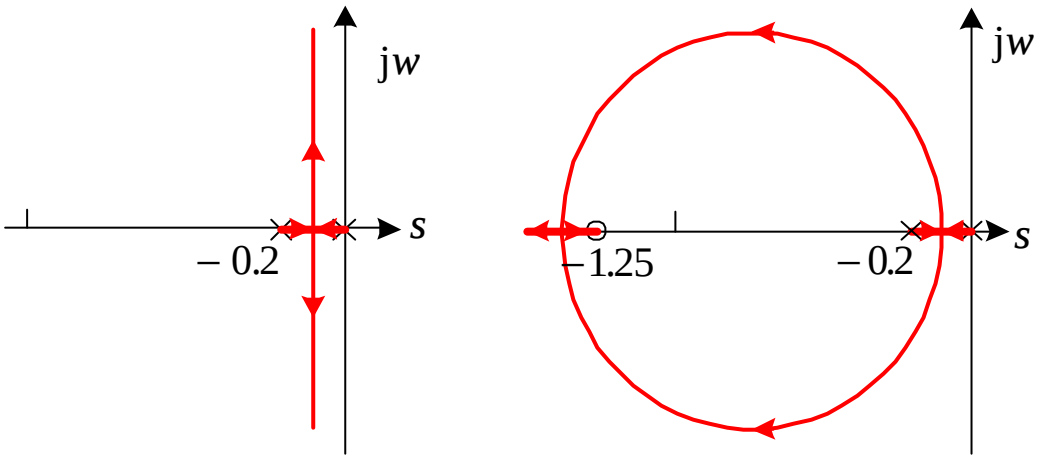
$$G_{\mathrm{II}}(s)F_{\mathrm{II}}(s)=\frac{5(1+0.8s)}{s(5s+1)}=\frac{0.8(s+1.25)}{s(s+0.2)}$$

- **Velocity F/B**



$$G_{III}(s)F_{III}(s) = \frac{5(1+0.8s)}{s(5s+1)} = \frac{0.8(s+1.25)}{s(s+0.2)}$$

(ii) Root Loci



Remarks:

System I $s_{1,2} = -0.1 \pm j0.995,\ z = 0.1$

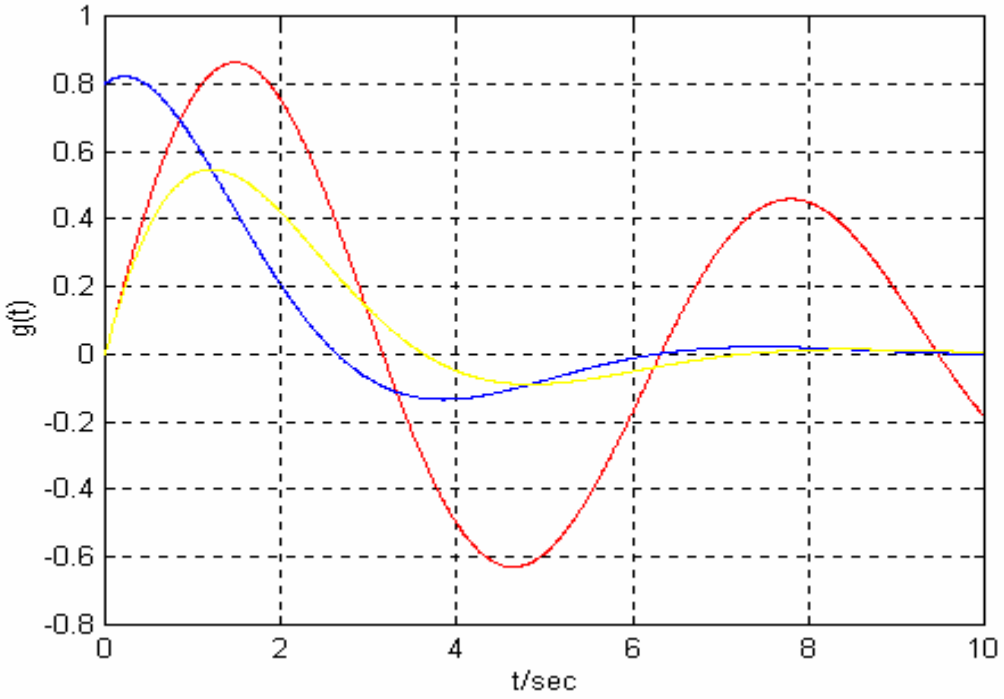
Heavy oscillation, slow decaying rate

System II,III $s_{1,2} = -0.5 \pm j0.866,\ z = 0.5$

Better performance could be expected

(iii) Time response

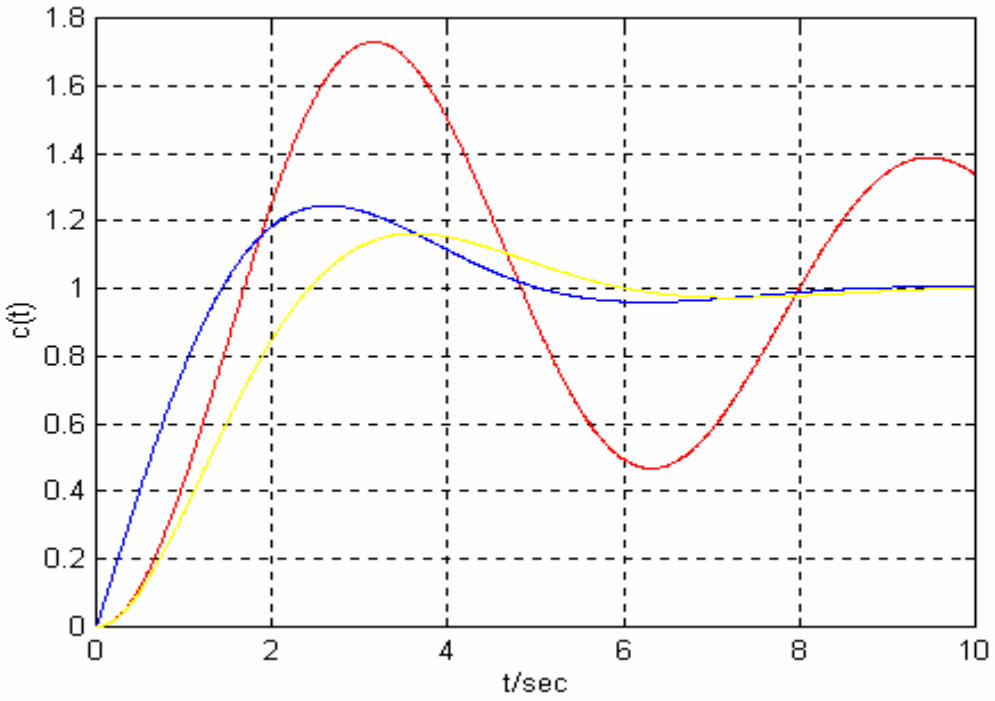
- **Unit impulse response**



★**The zero causes the difference between systems II and III**

(See Appendix V for impulse response equations)

●Unit step response



✶ II derivative of position error,
fast response.

III velocity F/B,
correction before error occurs,
smaller overshoot.

(See Appendix V for step response equations)

4.3 Conditionally Stable Systems

Example 4.5 Given OL transfer function

$$G(s)=\frac{K(s^2+2s+4)}{s(s+4)(s+6)(s^2+1.4s+1)}$$

with $F(s)=1$**. Find the stability ranges of gain** K**.**

Solution:

(i) OL poles: $0,\ -4,\ -6,\ -0.7\pm \mathrm{j}0.714$

OL zeros: $-1\pm \mathrm{j}1.7321$

Breakaway point: -2.3557

(ii) Imag axis crossing points:

CL char equation:

$$f(s)=s^5+11.4s^4+39s^3+(43.6+k)s^2$$
$$+(24+2K)s+4K=0$$

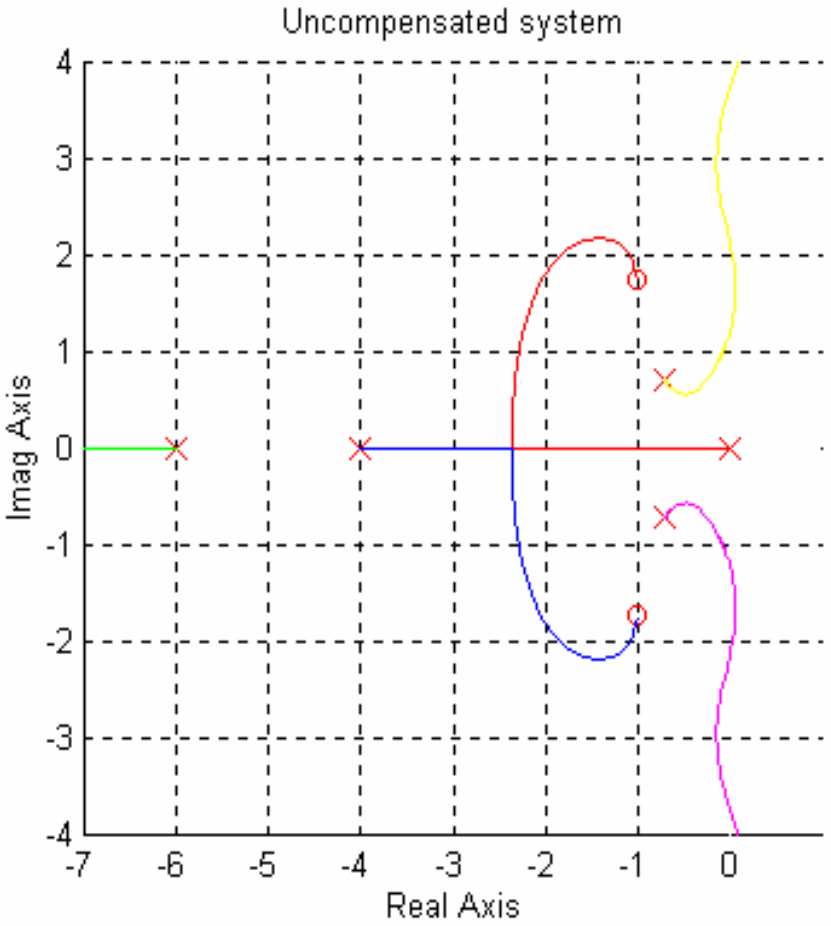
$$\begin{cases}11.4w^4-(43.6+K)w^2+4K=0 & (1)\\ w[w^4-39w^2+(24+2K)]=0 & (2)\end{cases}$$

$$K = -0.5w^4 + 19.5w^2 - 12 \qquad \text{(from (2))}$$

$$w^6 - 20.2w^4 + 92.9w^2 - 96 = 0$$

$$w_1 = 1.2115 \quad w_2 = 2.1545 \quad w_3 = 3.7538$$

with $K_1 = 15.54 \quad K_2 = 64.74 \quad K_3 = 163.51$

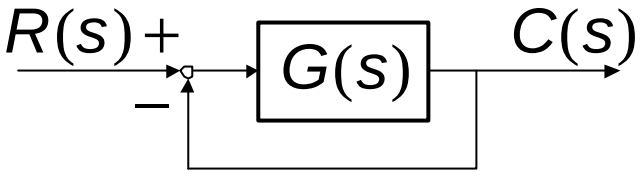


(iii) Stability ranges

$$K < 15.54, \quad 64.74 < K < 163.51$$

4.4 Non-minimum Phase Systems and Positive Feedback

(1) Root Locus Condition for Non-minimum Phase Systems



where $G(s)=\dfrac{K(1-T_1 s)}{s(1+Ts)}, T_1>0, K>0, T>0$

Angle condition:

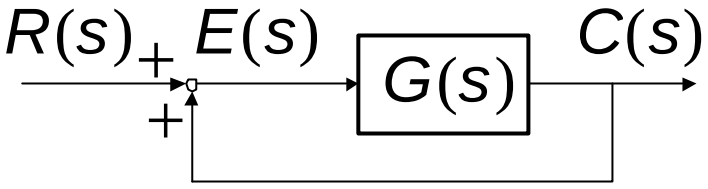
$$\arg G(s)=\arg\frac{K(1-T_1 s)}{s(1+Ts)}$$

$$=\arg\left[-\frac{K(T_1 s-1)}{s(Ts+1)}\right]$$

$$=180°+\arg\left[\frac{K(T_1 s-1)}{s(Ts+1)}\right]$$

So $\arg\dfrac{K(T_1 s-1)}{s(Ts+1)}=\pm 360°\times k,\ \ k=0,1,2,\cdots$

(2) Root Locus Condition for a System Under Positive Feedback



$$C(s) = G(s)E(s) = G(s)[R(s) + C(s)]$$

$$= G(s)R(s) + G(s)C(s)$$

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)}$$

Char eqn of CL system: $1 - G(s) = 0$

$$G(s) = 1$$

Angle condition:

$$\arg G(s) = \pm 360^\circ \times k, \qquad k = 0,1,2,\cdots$$

(3) Complementary Root Loci and Complete Root Loci

Let $G(s)F(s) = \dfrac{K(s-z_1)\cdots(s-z_m)}{(s-p_1)(s-p_2)\cdots(s-p_n)}$

$0 < K < \infty$: **Root loci**

$-\infty < K < 0$: **Complementary root loci**

$-\infty < K < \infty$: **Complete root loci**

(4) Plotting of Complementary Root Loci for Negative Feedback with $K < 0$

Char eqn $G(s)F(s) = KW(s) = -1$ **(**$K < 0$**)**

Let $K' = -K$**, then**

$K'W(s) = 1$ **(**$K' > 0$**)**

Root locus conditions:

$$\begin{cases} |K'W(s)| = 1 \\ \arg K'W(s) = \pm 360° \times k, \quad k = 0,1,2,\cdots \end{cases}$$

★All rules related to the angle condition should be modified.

- **Real axis locus**



Even number of real poles/zeros

- **Asymptotes**

$$g = \frac{\pm 360° \times k}{n-m}, \quad k = 0,1,2,\cdots,n-m-1$$

- **Angle of departure and arrival**

$$j_{pr} = \pm 360° \times k + \sum_{i=1}^{m} \arg(p_r - z_i) - \sum_{\substack{j=1 \\ j \neq r}}^{n} \arg(p_r - p_j)$$

$$j_{zr} = \pm 360° \times k - \sum_{\substack{i=1 \\ i \neq r}}^{m} \arg(z_r - z_i) + \sum_{j=1}^{n} \arg(z_r - p_j)$$

Example 4.6 Sketch root locus plot for

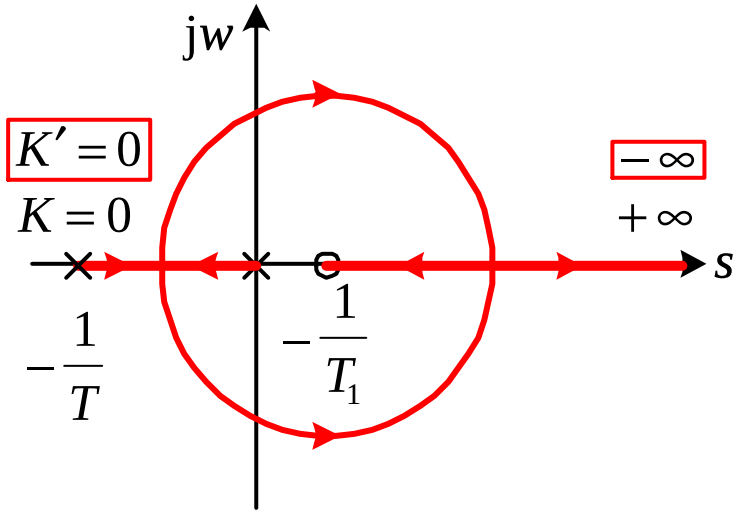
$$G(s)F(s)=\frac{K(1-T_1s)}{s(Ts+1)},\quad K>0,T_1>0,T>0$$

Solution: Write

$$G(s)F(s)=\frac{-K(T_1s-1)}{s(Ts+1)}=\frac{K'\left(s-\dfrac{1}{T_1}\right)}{s\left(s+\dfrac{1}{T}\right)}$$

where $K'=\dfrac{-KT_1}{T}<0$

★The root locus for $0<K<\infty$ **is the complementary root locus for** $-\infty<K'<0$**.**

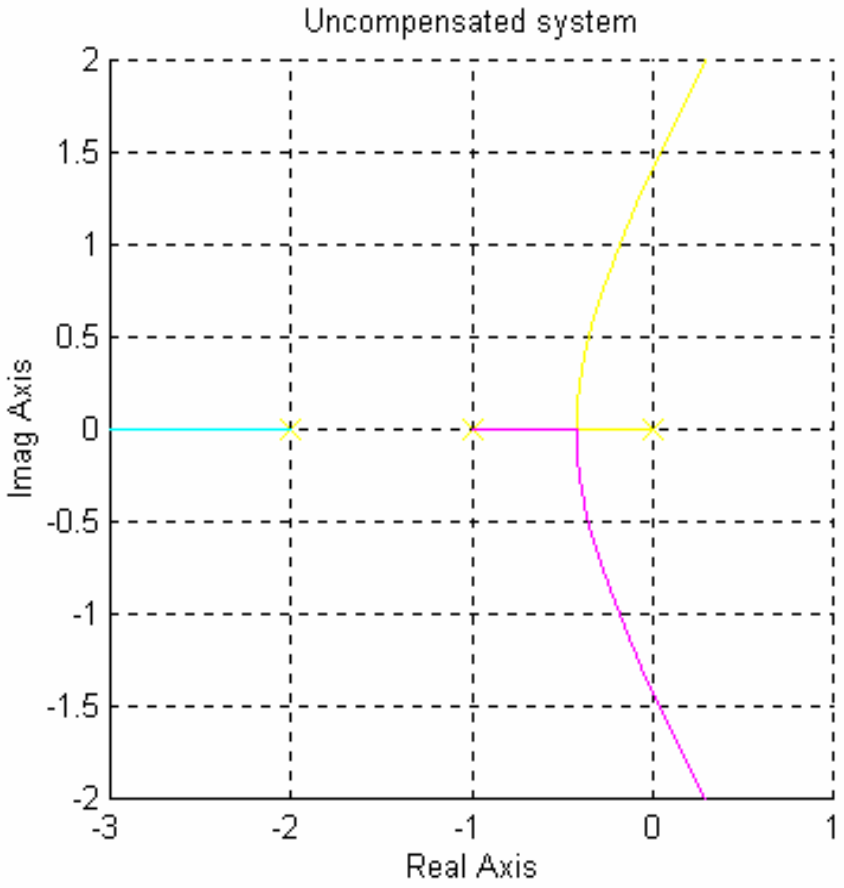


□

Example 4.7 Sketch the complete root locus plot for given system

$$G(s) = \frac{K}{s(s+1)(s+2)}, \quad F(s) = 1$$

Solution:



□

4.5 Systems with Pure Time Delay

$$G(s)F(s) = KW(s)\mathrm{e}^{-Ts}$$

where $W(s)$ is a rational function.

Let $s = s + \mathrm{j}w$, then $\mathrm{e}^{-Ts} = \mathrm{e}^{-Ts - \mathrm{j}Tw}$

CL characteristic equation:

$$1 + KW(s)\mathrm{e}^{-Ts} = 0$$

i.e. $KW(s)\mathrm{e}^{-Ts - \mathrm{j}Tw} = -1$

Since

$$\begin{cases} |G(s)F(s)| = |KW(s)|\ \mathrm{e}^{-Ts} \\ \arg[G(s)F(s)] = \arg[KW(s)] - Tw \end{cases}$$

We have root locus conditions

$$\begin{cases} |KW(s)|\ \mathrm{e}^{-Ts} = 1 \\ \arg W(s) = \pm p\,(2k+1) + wT \end{cases}$$

$$= \pm 180°\,(2k+1) + 57.3° \times wT$$

$$k = 0,1,2\cdots$$

(1) Starting points $(K = 0)$

$$\lim_{K\to 0} |W(s)| e^{-Ts} = \lim_{K\to 0} \frac{1}{K} = \infty$$

★**All poles of** $W(s)$ **and** $s \to -\infty$

(2) Terminating points $(K \to \infty)$

$$\lim_{K\to\infty} |W(s)e^{-Ts}| = \lim_{K\to\infty} \frac{1}{K} = 0$$

★**All zeros of** $W(s)$ **and** $s \to +\infty$

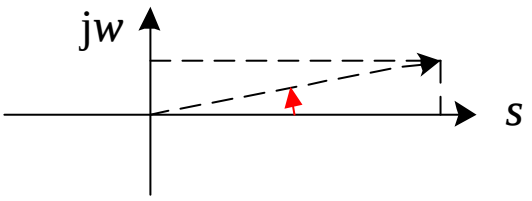
(3) Asymptotes

★$s \to \infty$ **when** $K \to 0$ **and** $K \to \infty$

$$\arg W(s)\big|_{s\to\infty} = \arg \frac{1}{s^{n-m}}$$

$$= -(n-m)\arctan\frac{w}{s}$$

● $s \to +\infty$



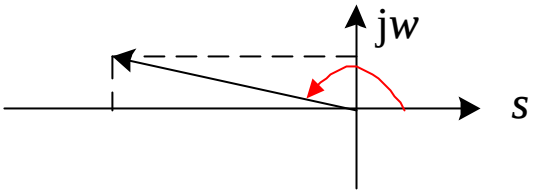
$$\arctan\frac{w}{s} \to 0$$

$$\arg W(s) = \pm p(2k+1) + wT = 0$$

$$w = \frac{\pm p(2k+1)}{T} = \frac{\pm 180°(2k+1)}{57.3T}$$

$$k = 0,1,2,\cdots$$

● $s \to -\infty$



$$\arctan\frac{w}{s} \to p$$

$$\arg W(s) = \pm p(2k+1) + wT = -(n-m)p$$

$$w = -\frac{(n-m) \pm (2k+1)}{T}p$$

★**If** $n-m$ **is odd**

$$w = \frac{2kp}{T}, \quad k = \pm 0, \pm 1, \pm 2, \cdots$$

★**If** $n-m$ **is even**

$$w = \frac{(2k+1)p}{T}, \quad k = \pm 0, \pm 1, \pm 2, \cdots$$

★**All asymptotes are parallel to the real axis**

(4) Number of branches: infinite

(5) Real axis root locus

★**Same as that of** $G(s)F(s) = KW(s)$ **since** $w = 0$

(6) Breakaway points

$$\frac{\mathrm{d}K}{\mathrm{d}s}=0, \quad \frac{\mathrm{d}}{\mathrm{d}s}\left[-\frac{\mathrm{e}^{Ts}}{W(s)}\right]=0$$

Example 4.8 Sketch the root locus plot for

$$G(s)=\frac{K\mathrm{e}^{-Ts}}{s+1}, \ F(s)=1, \ T=1$$

Solution:

(i) Write $W(s)=\dfrac{1}{s+1}$, $K=-(s+1)\mathrm{e}^{Ts}$

(ii) Asymptotes:

$s\to+\infty$: $w=\pm p,\pm 3p,\pm 5p,\cdots$

$s\to-\infty$: n-m=1 **is odd**

$$w=0,\pm 2p,\pm 4p,\pm 6p,\cdots$$

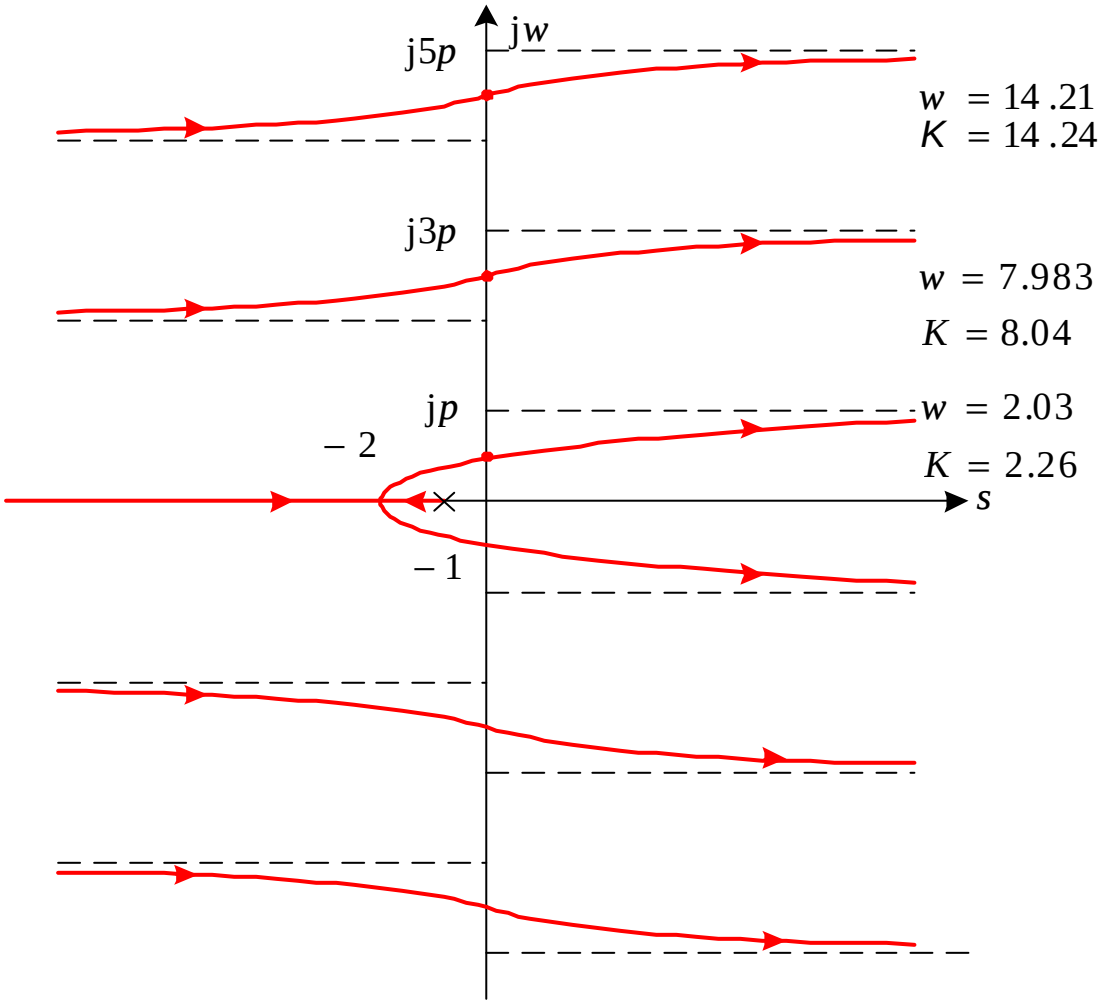
(iii) Real axis locus: $(-\infty, -1)$

(iv) Breakaway point

$$\frac{\mathbf{d}}{\mathbf{d}s}\left[-(s+1)\mathbf{e}^{Ts}\right] = -T(s+1)\mathbf{e}^{Ts} - \mathbf{e}^{Ts}$$

$$= \mathbf{e}^{Ts}(-Ts - T - 1) = 0$$

$$s = -\frac{T+1}{T} = -2$$



□

★Delay causes instability.

(See Appendix VI for detailed calculations)

4.6 Root Contour

Characteristic equation is

$$P(s)+K_1Q_1(s)+K_2Q_2(s)=0$$

where $P(s),Q_1(s),Q_2(s)$ **are polynomials.**

(1) Let $K_2=0$**, then** $P(s)+K_1Q_1(s)=0$

$$1+\frac{K_1Q_1(s)}{P(s)}=0$$

corresponding OL TF

$$G_1(s)F_1(s)=\frac{K_1Q_1(s)}{P(s)}$$

★**The roots of** $P(s)+K_1Q_1(s)=0$ **are on the root locus for** $K_1=0\rightarrow\infty$

(2) Write the char eqn as

$$1+\frac{K_2Q_2(s)}{P(s)+K_1Q_1(s)}=0$$

corresponding OL TF

$$G_2(s)F_2(s) = \frac{K_2 Q_2(s)}{P(s) + K_1 Q_1(s)}$$

★OL poles of $G_2(s)F_2(s)$ **are on the root locus of** $G_1(s)F_1(s)$

★Root locus of $G_2(s)F_2(s)$ **starts from the root locus of** $G_1(s)F_1(s)$ **for a given** K_1

★ A set of root loci is formed as K_1 **varies**

Example 4.9 Given

$$G(s) = \frac{K}{s(s+a)}, \quad F(s) = 1$$

Sketch the root contour for varying K **and** a**.**

Solution: CL char eqn:

$$s^2 + as + K = 0$$

(i) Let $a = 0$**, then** $s^2 + K = 0$

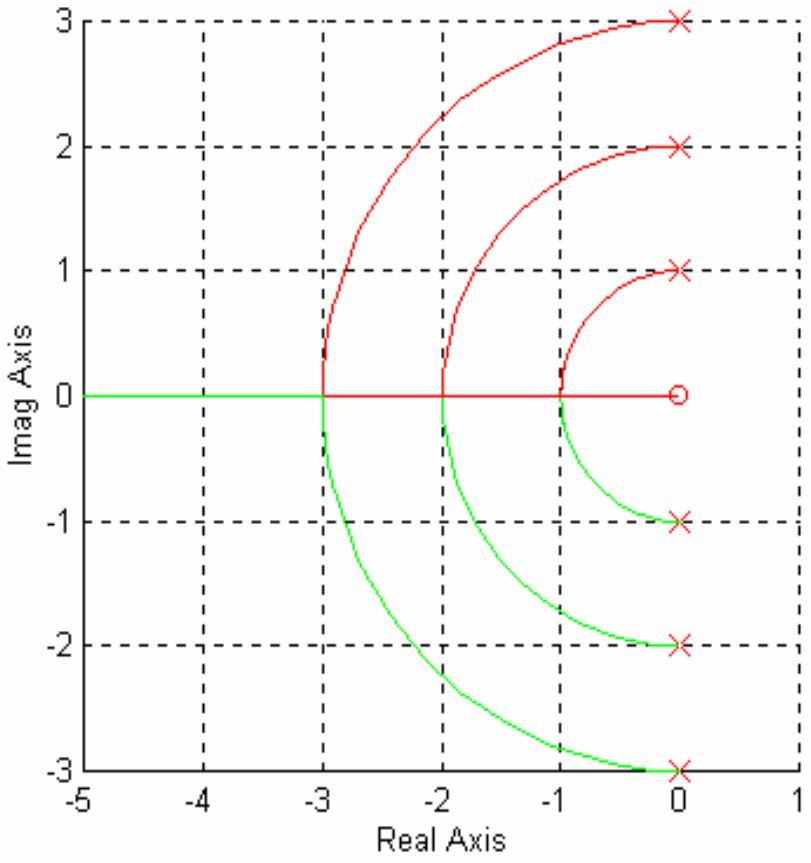
Sketch root locus of $G_1(s)F_1(s) = \dfrac{K}{s^2}$

(ii) Set $K = \text{const}$, we have

$$1 + \frac{as}{s^2 + K} = 0$$

Sketch the root locus of $G(s)F(s) = \dfrac{as}{s^2 + K}$

OL poles: $\pm j\sqrt{K}$



□

★Root locus for a varying parameter other than gain can be constructed in the similar way

4.7 Roots of a Polynomial

Example 4.10 To find approximately the roots of the polynomial

$$f(s) = 3s^4 + 10s^3 + 21s^2 + 24s - 16$$

Solution:

(i) Rewrite $f(s) = 0$ **as**

$$1 + \frac{24s - 16}{3s^4 + 10s^3 + 21s^2} = 0$$

so $G_1(s)F_1(s) = \dfrac{24s - 16}{3s^4 + 10s^3 + 21s^2}$

$$= \frac{8(3s - 2)}{s^2(3s^2 + 10s + 21)} = \frac{K\left(s - \frac{2}{3}\right)}{s^2\left(s^2 + \frac{10}{3}s + 7\right)}$$

(ii) Sketch root locus for $G_1(s)F_1(s)$**:**

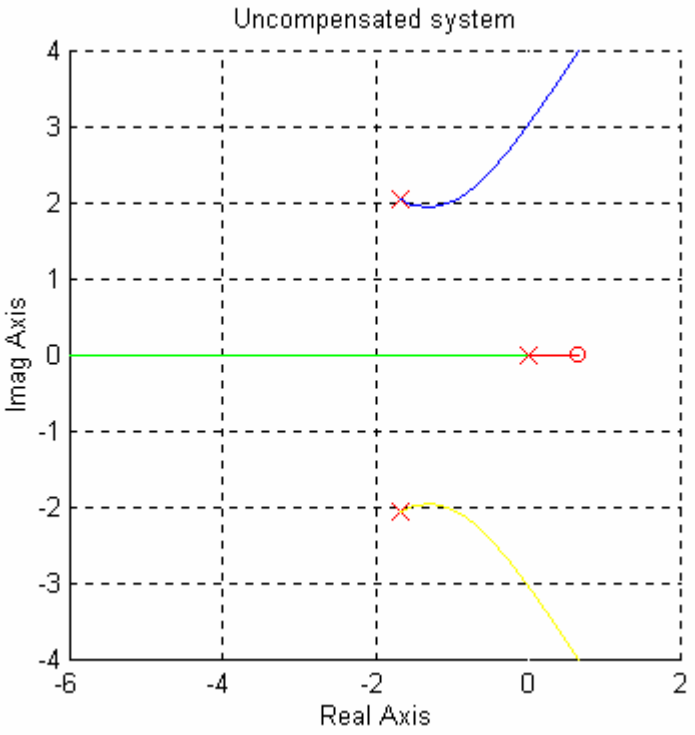
OL poles: $0,\ 0,\ -1.667 \pm \mathrm{j}2.055$

OL zeros: $2/3$

Asymptotes: $g = 60°, 180°, -60°,$

$$-s_a = -4/3$$

Imaginary axis crossing point: $\pm$ **j**3.037



(We may find by trial and error

$$s_1 = 0.4482, \quad s_2 = -2.2627,$$

and $s_{3,4} = -0.7459 \pm$ **j**2.1638**)**

(iii) Rewrite $f(s) = 0$ **as**

$$1 + \frac{21s^2 + 24s - 16}{3s^4 + 10s^3} = 0$$

so $G_2(s)F_2(s) = \dfrac{21s^2 + 24s - 16}{3s^4 + 10s^3}$

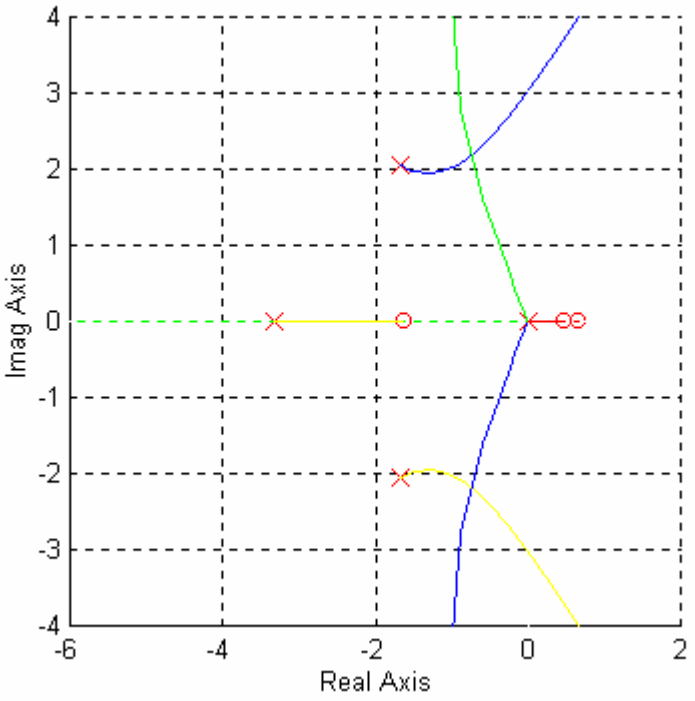
$$= \frac{7\left(s^2 + \frac{8}{7}s - \frac{16}{21}\right)}{s^3\left(s + \frac{10}{3}\right)}$$

(iv) Sketch root locus for $G_2(s)F_2(s)$

OL poles: $0,\ 0,\ 0,\ -10/3$

OL zeros: $0.4719,\ -1.6147$

$-s_a = -2.19$



★Crossing points of the two root locus plots give the roots of the polynomial. □